Comp HW7 P1

Three states: q_a, q_b, q_c, f, ~~AA~~, h

For reading as, bs, and cs

For each a, push A

For each b while As on the stack, remove an A

~~For~~ After that point, start pushing bs

For each c, remove a B

If we read $\epsilon, \perp$ in state q_c, $\rightarrow f$

If we read any symbol out of order, $\rightarrow h$

Start state, q_a

$(q_a, a, \perp) \rightarrow (q_a, A\perp)$

$(q_a, a, A)\$ \rightarrow (q_a, AA)$

// some As at the start

$(q_a, b, A) \rightarrow (q_b, \epsilon)$

$(q_b, b, A) \rightarrow (q_b, \epsilon)$

~~// B debt~~ B debt

$(q_a, b, \perp) \rightarrow (q_b, B\perp)$

$(q_b, b, \perp) \rightarrow (q_b, B\perp)$

$(q_b, b, B) \rightarrow (q_b, BB)$

// cs

$(q_b, c, B) \rightarrow (q_c, \epsilon)$

$(q_c, c, B) \rightarrow (q_c, \epsilon)$

$(q_c, \epsilon, \perp) \rightarrow (f, \epsilon)$

// too many cs

$(q_c, c, \perp) \to (h, \epsilon)$

// violates order

$(q_a, c, *) \to (h, \epsilon)$

$(q_b, a, *) \to (h, \epsilon)$

$(q_c, a, *) \to (h, \epsilon)$

$(q_c, b, *) \to (h, \epsilon)$

b) si q_a

$(q_a, a, \perp) \to (q_a, A\perp)$

$(q_a, b, A) \to (q_b, \epsilon)$

$(q_b, b, \perp) \to (q_b, B\perp)$

$(q_b, b, B) \to (q_b, BB)$

$(q_b, c, B) \to (q_c, \epsilon)$

$(q_c, c, B) \to (q_c, \epsilon)$

$(q_c, \epsilon, \perp) \to (f, \epsilon)$

Accept by final state

	$\perp$
a	$A\perp$
b	$\perp$
b	$B\perp$
b	$BB\perp$
c	$B\perp$
c	$\perp$

c) When the machine reads its first b, it transitions to state b. When it reads the second A after a B, it will ~~[illegible]~~ trigger $(q_b, a, *) \to (h, \epsilon)$. There is no way back from the halting state h. Thus the machine cannot accept.